

26. A sound wave travelling at a speed of 330 m/s produces 20 crests and 20 troughs in 0.1 second. The wavelength of the sound wave is :

(a) 1.1 m
(b) 3.3 m
(c) 1.65 m
(d) 2.2 m

$$\lambda = \frac{v}{f}$$

$v = 330 \text{ m/s}$
 $f = \frac{1}{t} = \frac{1}{0.1} = 10 \text{ Hz}$
 $\lambda = \frac{330}{10} = 33 \text{ m}$

27. An X-ray can be deflected :
- (a) by a magnetic field.
(b) by an electric field.
(c) by both electric and magnetic field.
(d) neither by electric field nor by magnetic field.

28. In a simple astronomical telescope, the objective and the eyepiece used respectively, are :

(a) a convergent lens and a divergent lens.
(b) a divergent lens and a divergent lens.
(c) a divergent lens and a convergent lens.
(d) a convergent lens and a convergent lens.

29. Twinkling of a star is due to :

(a) Interference of light
(b) Refraction of light
(c) Polarization of light
(d) Diffraction of light

30. Van de Graaff generator generates :

(a) electrostatic charge
(b) magnetic field
(c) high voltage low alternating current
(d) high voltage low direct current

31. Which one of the following metals is widely extracted by the electrolysis of its molten compound ?

(a) Cu
(b) Au
(c) Sn
(d) Na

32. Which one of the following is *not* a property of 'graphene' ?

(a) It is the thinnest material known so far.
(b) It is almost completely transparent.
(c) It is highly conducting.
(d) It is a wide band-gap semiconductor.

33. Why do plastics *not* degrade easily ?

(a) They have strong ionic bonds.
(b) They have strong covalent bonds.
(c) They have strong metallic bonds.
(d) They have very high melting points ($> 500^\circ\text{C}$).

34. Which one of the following is the correct chemical formula of carboxylic acid ?

(a) $\text{C}_3\text{H}_7\text{OH}$
(b) C_3H_8
(c) $\text{C}_2\text{H}_5\text{COOH}$
(d) CH_3COCH_3

35. Which one of the following statements is **not** correct for benzene ?
- Each carbon atom forms sigma bonds with two other carbon atoms with a bond angle of 120° .
 - ☒ Delocalized electrons create a symmetrical 'cloud' of electrons above and below the plane.
 - The length of C - C bonds is intermediate between single and double bonds.
 - It has six isomers.
36. Which one of the following is **not** a nitrogen fertilizer ?
- $(\text{NH}_4)_2\text{SO}_4$
 - NH_4NO_3
 - N_2
 - $(\text{NH}_2)_2\text{CO}$
37. Which one of the following general electronic configurations correctly represent a transition metal element ?
- ☒ $(n-2) d^{1-10} ns^2$
 - $(n-2) f^{1-14} (n-1) d^{0-1} ns^2$
 - $ns^2 np^6 nd^{1-10}$
 - $(n-1) d^{1-10} ns^{0-2}$
38. Variable oxidation numbers are possible for :
- Sodium
 - Calcium
 - ☒ Iron
 - Lithium
39. Which one of the following gas smells like a rotten egg ?
- ☒ Ammonia
 - Hydrogen sulphide
 - Acetylene
 - Sulphur dioxide
40. The chemical reaction
- $$\text{K}_2\text{SO}_4 + \text{BaCl}_2 \longrightarrow 2\text{KCl} + \text{BaSO}_4 \downarrow$$
- (aq.) (aq.) (aq.) (solid)
- is an example of :
- Addition reaction
 - Displacement reaction
 - Decomposition reaction
 - ☒ Double displacement reaction
41. Which one among the following is the correct scheme followed for classification of animals ?
- Class $\rightarrow$ Phylum $\rightarrow$ Order $\rightarrow$ Genus $\rightarrow$ Family $\rightarrow$ Species
 - Phylum $\rightarrow$ Class $\rightarrow$ Family $\rightarrow$ Genus $\rightarrow$ Order $\rightarrow$ Species
 - Order $\rightarrow$ Class $\rightarrow$ Phylum $\rightarrow$ Family $\rightarrow$ Species $\rightarrow$ Genus
 - ☒ Phylum $\rightarrow$ Class $\rightarrow$ Order $\rightarrow$ Family $\rightarrow$ Genus $\rightarrow$ Species
42. Which one of the following is an example of airborne disease ?
- ☒ Tuberculosis
 - Malaria
 - Dengue
 - Cholera